

Table 1: Comparative Overview of ICS/IoT Datasets Used in Anomaly Detection Research

Dataset	Year	Focus	Characteristics	Used For
Modbus-TCP	2023	ICS / SCADA Network Traffic	Wireshark packet captures of a simulated master-slave Modbus/TCP substation network; 44,536 packets (train: 37,469 test: 7,067); 7 features ; dominant protocols: Modbus/TCP, TCP, ARP; binary labels (normal / attack); attack type: SYN Flood DoS on port 502; highly imbalanced	Anomaly detection & binary classification on industrial Modbus/TCP traffic;
SWaT	2015	Water Treatment ICS	Physical 11-stage water treatment testbed; 54,621 records at 1-second granularity; 51 features; binary label (<i>Normal/Attack</i>); capture period: Dec. 2015 – Jan. 2016; dataset in this version contains only attack samples (used as the attack-phase split)	Anomaly detection; attack classification;
ICS-Flow	2022	ICS Network Flow (Modbus)	Bi-directional network flow dataset from a Modbus/TCP ICS testbed; 45,718 flows, 64 features ; protocols: IPv4-TCP ($\approx 96\%$), ARP, ICMP; dual labelling: IT perspective (30,236 normal 15,482 attack); NST perspective (36,706 normal 9,012 attack); 5 attack categories: replay, DDoS, port-scan, MitM, ip-scan; capture date: 2022-09-16 (≈ 2 hours)	Network intrusion detection for ICS; multi-class attack classification; flow-based anomaly detection
Electra	2022	ICS / SCADA – Electric Traction Substation (Railway)	Real-world Modbus/TCP traffic from a high-speed railway electric traction substation testbed; 16,289,277 records, 11 features; 4 network nodes (1 SCADA master + 3 slaves); dominant function code: FC3 (Read Holding Registers, $>99\%$); capture duration: ≈ 14 hours; fine-grained labelling: 7 attack categories; imbalanced: 85.3% normal vs. 14.7% attack;	ICS intrusion detection; multi-class Modbus attack classification;
BATADAL	2017	Cyberattacks detection in water distribution systems	Time-series SCADA data from water distribution system; hydraulic sensor readings; simulated false data injection and actuator attacks; labeled normal vs attack periods; widely used benchmark for ICS anomaly detection	Anomaly detection; deep learning (LSTM, autoencoders); ICS security research

Dataset	Year	Focus	Characteristics	Used For
TON_IoT	2020	IoT / IIoT / OT Cybersecurity	Realistic network traffic from IoT, IIoT and OT environments; 44 network features extracted using Zeek; multiple attack types (DoS, DDoS, backdoor, injection, ransomware, XSS, scanning); binary and multi-class labels	Intrusion Detection Systems (IDS); Federated Learning; anomaly detection; IoT/OT cybersecurity
X-IIoTID	2022	Extended IIoT (multi-layer IoT/OT)	Multi-layer IoT/IIoT dataset (fog, edge, cloud communication); 68 network features; $\approx 820,000+$ records; attacks include DDoS, injection, MITM, scanning; labeled normal vs anomaly and multi-class attack categories	Deep learning IDS; Federated Learning; multi-layer IoT/OT intrusion detection
IERAD	2024	Emergency response decision systems	Urban emergency response dataset (Sydney, 2018–2024); 24 features including traffic, weather, hospital capacity, drone/ambulance status; timestamped (10-minute intervals); labeled dispatch strategies (Drone, Ambulance, Hybrid)	AI decision systems; optimization; classification and routing models